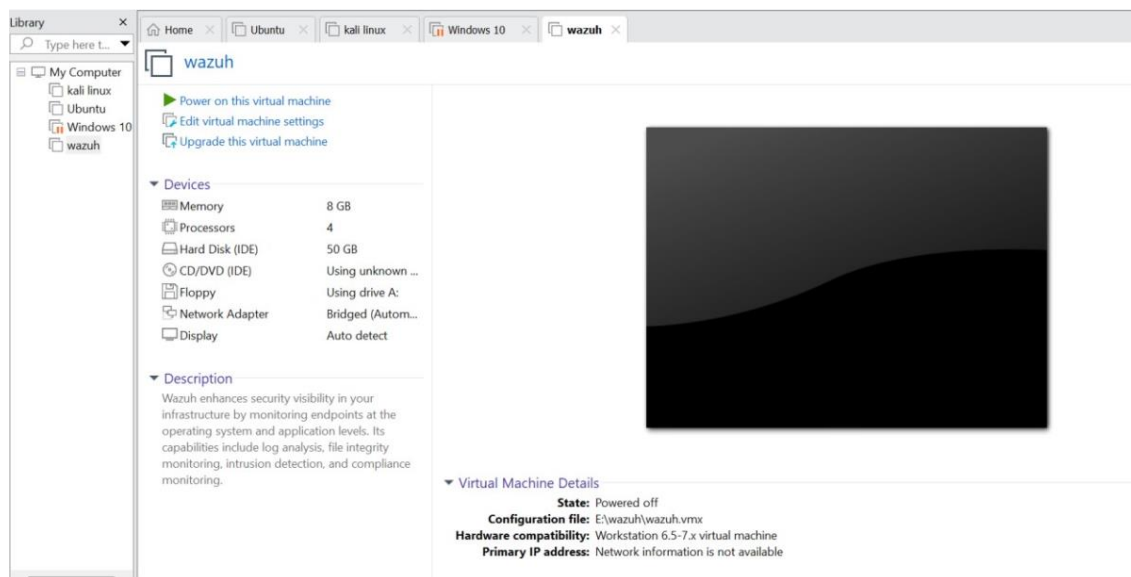
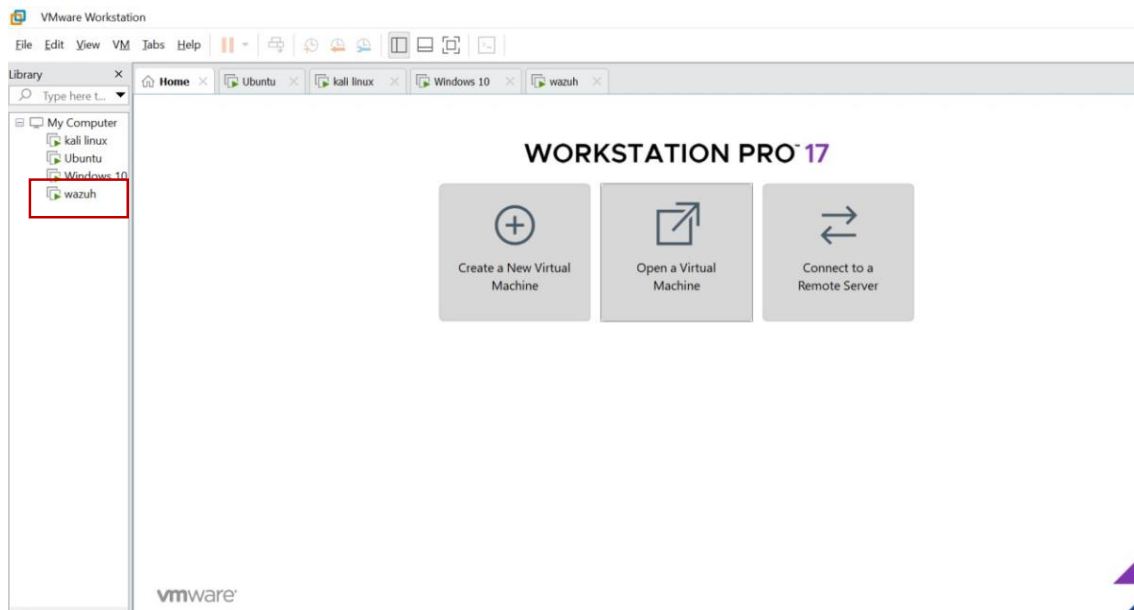


Name: Anees Ahmed

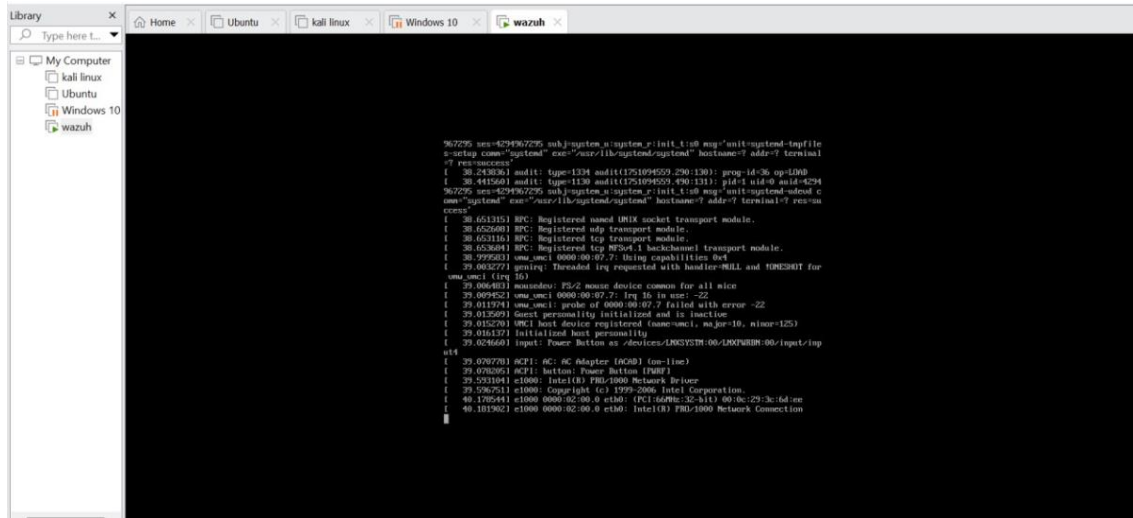
Role : Soc Analyst

Topic: Installation Of Wazuh and File Integrity Monitoring

1. Installation of Wazuh



1. Start Wazuh Machine

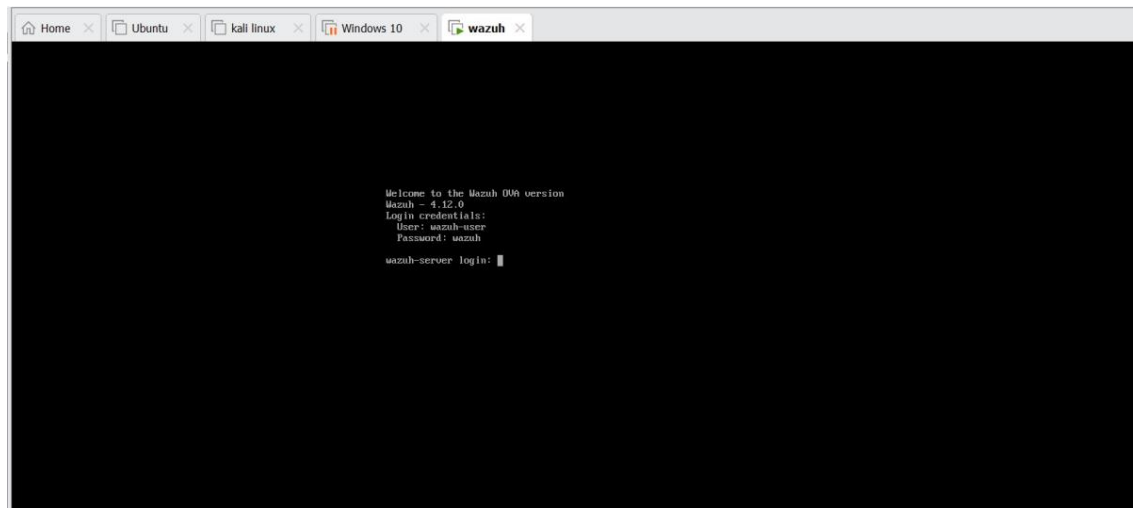


```
967275 ses-6294967295 wkhjrgtstet_a system_r init.1.108 wsg="init-systemd-tpmfile
s-setup.com" system="x86_64" lib/systemd/systemd" hostname="wazuh" add=7 terminal
? res=success"
I 38.2430261 audit: type=1334 audit(1751094559.390:130): prog-id=36 op=UNDE
I 38.4415601 audit: type=1130 audit(1751094559.490:131): pid=1 uid=0 auid=6294
967275 ses-6294967295 wkhjrgtstet_a system_r init.1.108 wsg="init-systemd-tpmfile
s-setup.com" lib/systemd/systemd" hostname="wazuh" add=7 terminal? res=su
cess"
I 38.6513151 IPC: Registered named UNIX socket transport module.
I 38.6526001 IPC: Registered udp transport module.
I 38.6531161 IPC: Registered tcp transport module.
I 38.6536941 IPC: Registered tcp NFSv4.1 backchannel transport module.
I 38.9995931 omc_omci 0000:00:07:7: Using capabilities 0x4
I 39.6035779 gpioirq: Threaded irq requested with handler NULL and FORCEIRQ for
omc_omci (irq 16)
I 39.0064013 watchdog: PS/2 mouse device common for all mice
I 39.0094521 omc_omci 0000:00:07:7: irq 16 in wsc1-22
I 39.0119741 omc_omci: probe of 0000:00:07:7 failed with error -22
I 39.0155091 Serial personality initialized and is inactive
I 39.0152791 OMCI host device registered (name=omci, major=10, minor=125)
I 39.0161191 initialized host personality
I 39.0246601 input: Power Button as devices/LNXSYSTM:00/LNXPWRBTN:00-input/inp
ut
I 39.0267201 ACPI: AC: AC Adapter (ACAD) (on-line)
I 39.0262051 ACPI: button: Power Button (PMBF)
I 39.5531041 c1000: Intel(R) PRO/1000 Network Adapter
I 39.5967511 c1000: Copyright (c) 1999-2006 Intel Corporation.
I 40.1705441 c1000 0000:02:00:0 e180 (PCI-e000:32-bit) 00:0c:27:3c:6d:ce
I 40.1817961 c1000 0000:02:00:0 e180 Intel(R) PRO/1000 Network Connection
```

2. Wazuh-user login Default login

User: wazuh-user

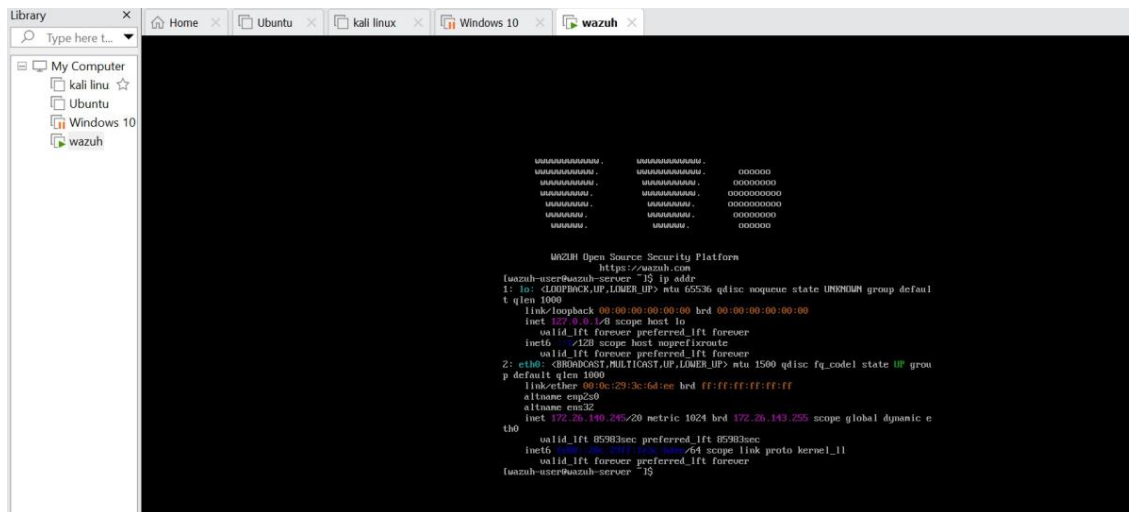
Password: wazuh



```
Welcome to the Wazuh OVA version
Wazuh - 4.12.0
Login credentials:
User: wazuh-user
Password: wazuh
wazuh-server login: █
```

3. Check Ip Address

Ip addr



```
WAZUH Open Source Security Platform
https://wazuh.com
wazuh-user@wazuh-server: ~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qlist: noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlist: fq_codel state UP grou
    p default qlen 1000
    link/ether 08:00:27:3c:6d:ee brd ff:ff:ff:ff:ff:ff
    altname ems20
    altname ems20
    inet 172.28.140.240/20 metric 1024 brd 172.28.143.255 scope global dynamic e
    th0
        valid_lft 85903sec preferred_lft 85903sec
        inet6 -::: scope link proto kernel_ll
    wazuh-user@wazuh-server: ~$
```

4. Check the systemctl commands for enabling, starting, and checking the status of all three core Wazuh components — Manager, Indexer, and Dashboard

1. Wazuh Manager

Enable Wazuh Manager to start on boot

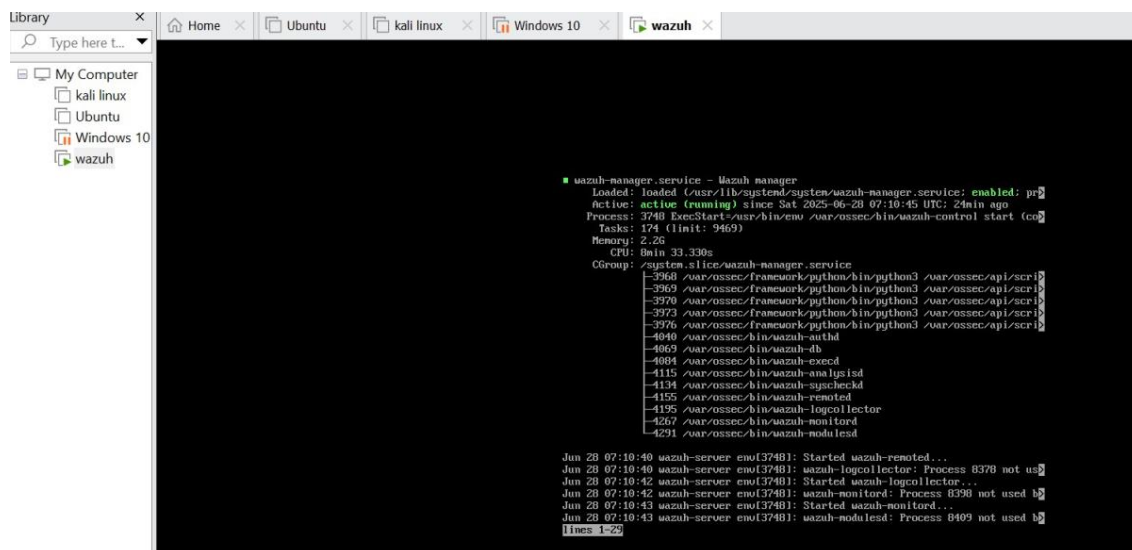
sudo systemctl enable wazuh-manager

Start Wazuh Manager

sudo systemctl start wazuh-manager

Check Wazuh Manager status

sudo systemctl status wazuh -manager



```
■ wazuh-manager.service - Wazuh manager
Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; pr
Active: active (running) since Sat 2025-06-28 07:10:45 UTC; 24min ago
Process: 3748 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co
Tasks: 174 (limit: 9469)
Memory: 2.2G
CPU: 8min 33.330s
CGroup: /system.slice/wazuh-manager.service
└─3968 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr
└─3969 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr
└─3970 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr
└─3973 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr
└─3976 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr
└─4040 /var/ossec/bin/wazuh-authd
└─4069 /var/ossec/bin/wazuh-db
└─4084 /var/ossec/bin/wazuh-execd
└─4115 /var/ossec/bin/wazuh-analysd
└─4134 /var/ossec/bin/wazuh-syscheckd
└─4155 /var/ossec/bin/wazuh-remoted
└─4195 /var/ossec/bin/wazuh-logcollector
└─4267 /var/ossec/bin/wazuh-monitord
└─4291 /var/ossec/bin/wazuh-modulesd

Jun 28 07:10:40 wazuh-server emul37481: Started wazuh-remoted...
Jun 28 07:10:40 wazuh-server emul37481: wazuh-logcollector: Process 8378 not us
Jun 28 07:10:42 wazuh-server emul37481: Started wazuh-logcollector...
Jun 28 07:10:42 wazuh-server emul37481: wazuh-monitord: Process 8398 not used b
Jun 28 07:10:43 wazuh-server emul37481: Started wazuh-monitord...
Jun 28 07:10:43 wazuh-server emul37481: wazuh-modulesd: Process 8409 not used b
Times 1=25
```

Enable Wazuh Indexer to start on boot

Start Wazuh Indexer

Check Wazuh Indexer status

```
Library
Type here t...

My Computer
kali linux
Ubuntu
Windows 10
wazuh

altname ems32
inet 172.28.149.245/20 metric 1024 brd 172.28.143.255 scope global dynamic e
th0
    valid_lft 65983sec preferred_lft 65963sec
    inet6 :::::/64 scope link proto kernel_l1
    valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$ sudo systemctl status wazuh-indexer
■ wazuh-indexer.service - wazuh-indexer
Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; pr
Active: active (running) since Sat 2025-06-28 07:12:05 UTC; 22min ago
Docs: https://documentation.wazuh.com
Main PID: 3747 (java)
Tasks: 86 (limit: 9469)
Memory: 4.5G
CPU: 3min 40.174s
CGroup: /system.slice/wazuh-indexer.service
└─3747 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensea

Jun 28 07:10:56 wazuh-server systemd-entrypoint[3747]: WARNING: A terminal de
Jun 28 07:10:56 wazuh-server systemd-entrypoint[3747]: WARNING: System: setSecu
Jun 28 07:10:56 wazuh-server systemd-entrypoint[3747]: WARNING: Please consid
Jun 28 07:10:56 wazuh-server systemd-entrypoint[3747]: WARNING: System: setSecu
Jun 28 07:11:00 wazuh-server systemd-entrypoint[3747]: Jun 28, 2025 7:11:00 AM
Jun 28 07:11:00 wazuh-server systemd-entrypoint[3747]: WARNING: OUTPUT locale p
Jun 28 07:11:03 wazuh-server systemd-entrypoint[3747]: WARNING: A terminal de
Jun 28 07:11:03 wazuh-server systemd-entrypoint[3747]: WARNING: System: setSecu
Jun 28 07:11:03 wazuh-server systemd-entrypoint[3747]: WARNING: Please consid
Jun 28 07:11:03 wazuh-server systemd-entrypoint[3747]: WARNING: System: setSecu

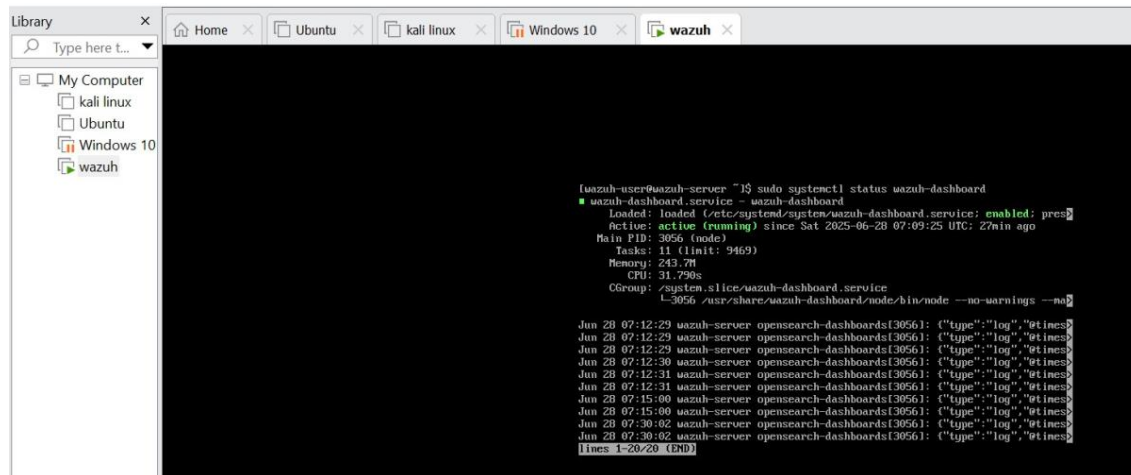
[wazuh-user@wazuh-server ~]$
```

Enable Wazuh Dashboard to start on boot

Start Wazuh Dashboard

Check Wazuh Dashboard status

```
sudo systemctl status wazuh-dashboard
```

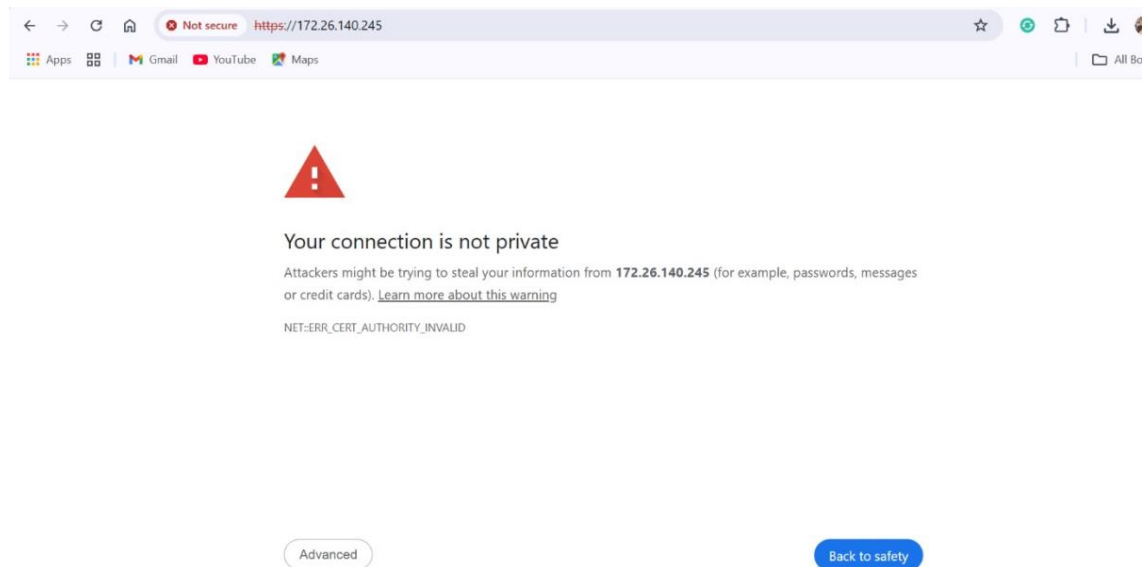


A terminal window titled 'wazuh' showing the command `sudo systemctl status wazuh-dashboard` and its output. The output indicates the service is loaded and active (running). Below the status, a series of log entries are shown, all of which are of type 'log' and timestamped 'Jun 28 07:12:29'.

```
[wazuh-user@wazuh-server ~]$ sudo systemctl status wazuh-dashboard
■ wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset:
   Active: active (running) since Sat 2025-06-28 07:09:25 UTC; 27min ago
     Main PID: 3056 (node)
       Tasks: 11 (limit: 9469)
      Memory: 243.7M
         CPU: 31.790s
    CGroup: /system.slice/wazuh-dashboard.service
           └─3056 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --no

Jun 28 07:12:29 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:12:29 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:12:29 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:12:30 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:12:31 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:12:31 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:15:00 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:15:00 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:30:02 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
Jun 28 07:30:02 wazuh-server opensearch-dashboards[3056]: {"type":"log","@tines
lines 1-20/20 (END)
```

5. Open Browser Types Wazuh Ip <https://172.26.140.245/>

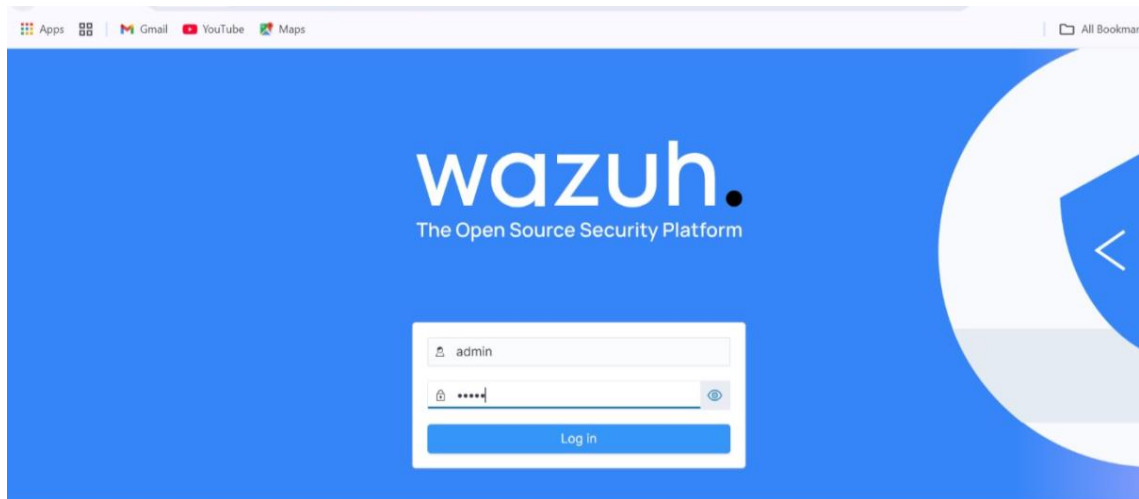


6. Login wazuh Dashboard

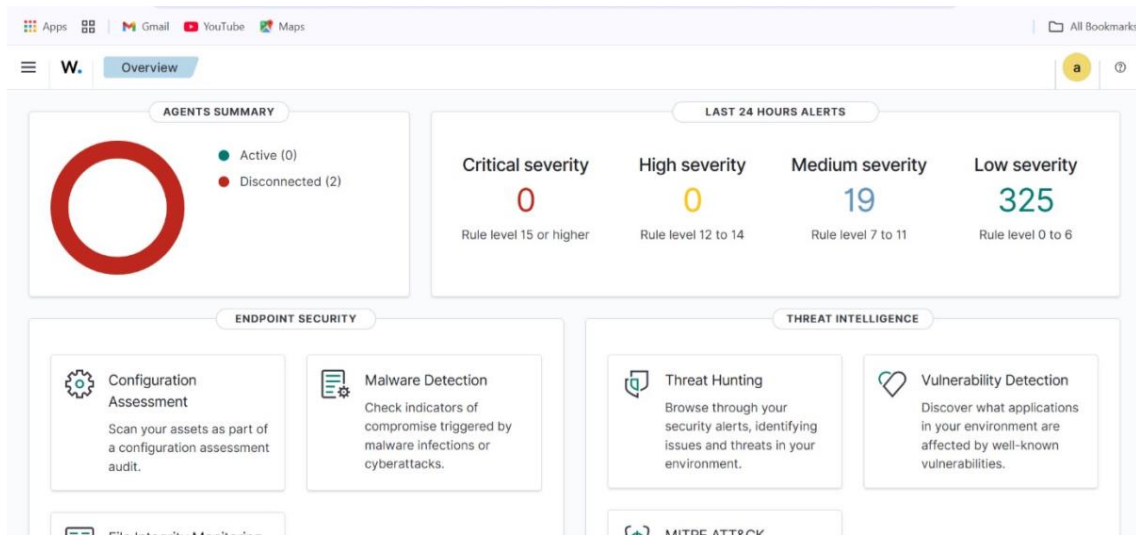
Default Login

User:Admin

Password:Admin



Wazuh Web Interface



7. Deploy wazuh-agent

← → ↻ 🔍 Not secure https://172.26.140.245/app/endpoints-summary#/agents-preview/deploy ☆ 📄 📁 All Bookmarks

Apps Gmail YouTube Maps

W. Endpoints Deploy new agent a ⓘ

Deploy new agent

✓ Select the package to download and install on your system:

 **LINUX**
☐ RPM amd64 ☐ RPM aarch64
☐ DEB amd64 ☐ DEB aarch64

 **WINDOWS**
☒ MSI 32/64 bits

 **macOS**
☐ Intel
☐ Apple silicon

📘 For additional systems and architectures, please check our [documentation](#).

2 Server address:

← → ↻ 🔍 Not secure https://172.26.140.245/app/endpoints-summary#/agents-preview/deploy ☆ 📄 📁 All Bookmarks

W. Endpoints Deploy new agent a ⓘ

DEB amd64 DEB aarch64

📘 For additional systems and architectures, please check our [documentation](#).

✓ Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ⓘ

172.26.140.245

☐ Remember server address

← → ↻ 🔍 Not secure https://172.26.140.245/app/endpoints-summary#/agents-preview/deploy ☆ 📄 📁 All Bookmarks

W. Endpoints Deploy new agent a ⓘ

172.26.140.245

☐ Remember server address

✓ Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name ⓘ

WindowSOC1

📘 The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups ⓘ

Default

After This Open powershell as administrator Paste following CMD

← → ↻ 🏠 🔒 Not secure https://172.26.140.245/app/endpoints-summary#/agents-preview/deploy ☆ 📄 🧑

Apps 🗄️ Gmail YouTube Maps 📁 All Bookmarks

☰ W. Endpoints Deploy new agent a ⓘ

Default ▾

4 Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.12.0-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='172.26.140.245' WAZUH_AGENT_NAME='WIndowSOC1'
```

④ Requirements

- You will need administrator privileges to perform this installation.
- PowerShell 3.0 or greater is required.

Keep in mind you need to run this command in a Windows PowerShell terminal.

← → ↻ 🏠 🔒 Not secure https://172.26.140.245/app/endpoints-summary#/agents-preview/deploy ☆ 📄 🧑

Apps 🗄️ Gmail YouTube Maps 📁 All Bookmarks

☰ W. Endpoints Deploy new agent a ⓘ

agent /q WAZUH_MANAGER='172.26.140.245' WAZUH_AGENT_NAME='WIndowSOC1'

④ Requirements

- You will need administrator privileges to perform this installation.
- PowerShell 3.0 or greater is required.

Keep in mind you need to run this command in a Windows PowerShell terminal.

5 Start the agent:

```
NET START WazuhSvc
```

Wazuh Services Started Successful

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Writing web request
Writing request stream... (Number of bytes written: 3056860)

agent /q WAZUH_MANAGER='172.26.140.245' WAZUH_AGENT_NAME='WIndowSOC1'
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

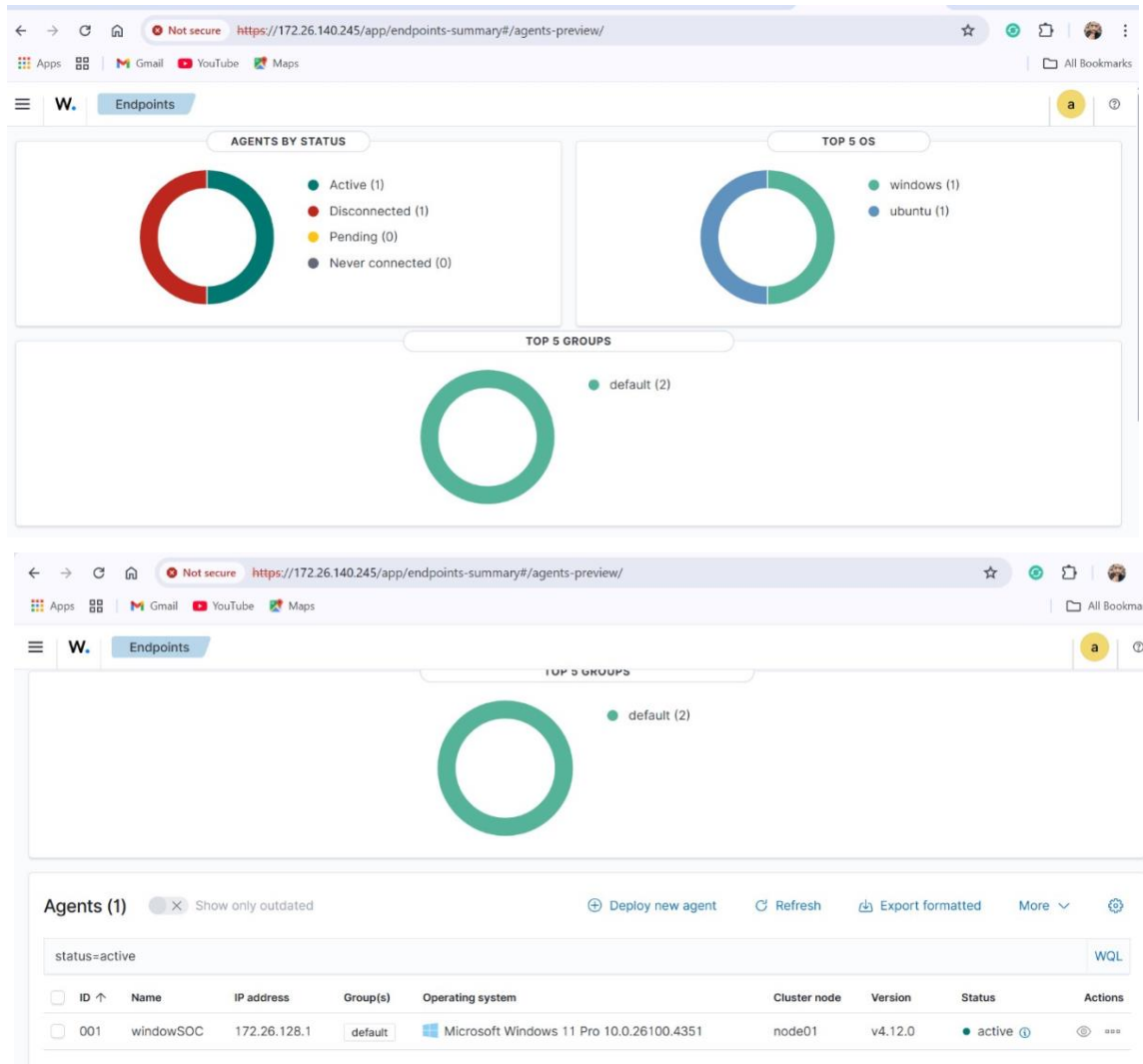
PS C:\WINDOWS\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.12.0-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='172.26.140.245' WAZUH_AGENT_NAME='WIndowSOC1'
PS C:\WINDOWS\system32> NET START WazuhSvc

The Wazuh service was started successfully.

PS C:\WINDOWS\system32>
```


After Started Refresh Browser

And window agent Deploy Successfully



8. Now Configure FIM to monitor critical files and directories.

Generate and tune alerts to detect unauthorized file changes

Access ossec.conf file in window

ossec-agent

C:\Program Files (x86)\ossec-agent

C:\Users\...AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup

Name	Date modified	Type	Size
active-response	26/06/2025 6:59 pm	File folder	
bookmarks	26/06/2025 6:59 pm	File folder	
incoming	26/06/2025 6:59 pm	File folder	
logs	27/06/2025 12:00 am	File folder	
queue	26/06/2025 6:59 pm	File folder	
rids	26/06/2025 7:00 pm	File folder	
ruleset	26/06/2025 6:59 pm	File folder	
shared	26/06/2025 7:00 pm	File folder	
syscheck	26/06/2025 6:59 pm	File folder	
tmp	28/06/2025 1:03 pm	File folder	
upgrade	23/06/2025 3:28 pm	File folder	
wodles	26/06/2025 6:59 pm	File folder	
.agent_info	28/06/2025 1:03 pm	AGENT_INFO File	1 KB
agent-auth	30/04/2025 11:05 am	Application	339 KB
agent-auth.exe.manifest	30/04/2025 10:48 am	MANIFEST File	3 KB
client.keys	26/06/2025 7:00 pm	KEYS File	1 KB

ossec-agent

This PC > Local Disk (C:) > Program Files (x86) > ossec-agent

Search ossec-ager

Name	Date modified	Type	Size
local_internal_options.conf.save	30/04/2025 10:52 am	SAVE File	1 KB
manage_agents			341 KB
ossec.conf			7 KB
ossec.conf.save			10 KB
ossec			70 KB
profile-10.template			1 KB
rsync.dll			330 KB
syscollector.dll			471 KB
sysinfo.dll			547 KB
VERSION.json			1 KB
vista_sec			92 KB
wazuh-agent	30/04/2025 11:05 am	Application	1,172 KB
wazuh-agent.state	28/06/2025 1:15 pm	STATE File	1 KB
wazuh-logcollector.state	28/06/2025 1:14 pm	STATE File	2 KB
win32ui	30/04/2025 11:05 am	Application	291 KB
win32ui.exe.manifest	30/04/2025 10:48 am	MANIFEST File	1 KB
wpk_root.pem	30/04/2025 10:48 am	PEM File	2 KB

Wazuh Agent

Manage View Help

Wazuh v4.12.0

Agent: windowSOC [001] - any

Status: Running

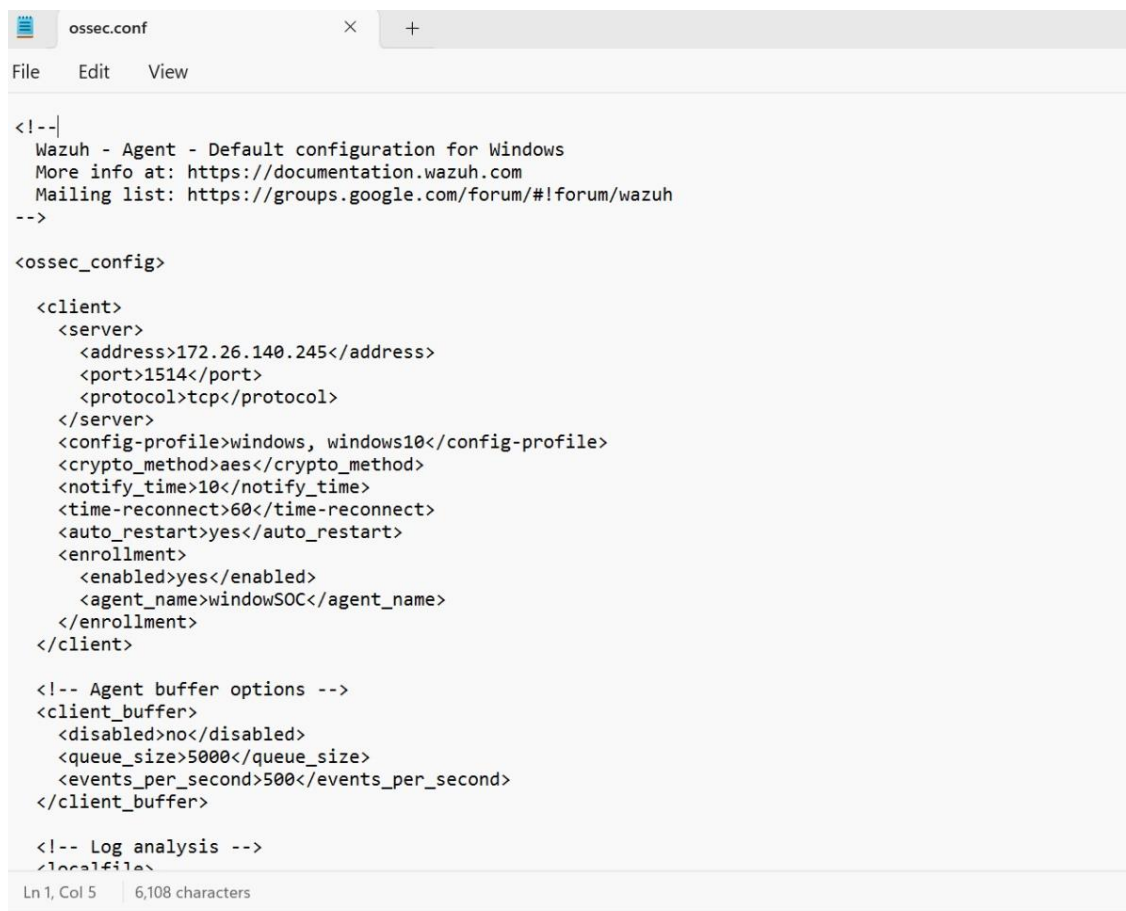
Manager IP: 172.26.140.245

Authentication key: MDAXlHdpbnRvd1NfQyBhbnk

Save Refresh

https://wazuh.com Revision rc1

Go View Option and open ossec.conf file



```
ossec.conf
File Edit View

<!--|
Wazuh - Agent - Default configuration for Windows
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>

  <client>
    <server>
      <address>172.26.140.245</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>windows, windows10</config-profile>
    <crypto_method>aes</crypto_method>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <enrollment>
      <enabled>yes</enabled>
      <agent_name>windowSOC</agent_name>
    </enrollment>
  </client>

  <!-- Agent buffer options -->
  <client_buffer>
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Log analysis -->
  <localfiles>
```

9. Configure Ossec.conf file

```
<!-- File integrity monitoring -->
<syscheck>
  <enabled>yes</enabled>

  <!-- Directories to be monitored -->
  <directories realtime="yes">C:\Users\hbk72</directories>
  <directories realtime="yes">C:\Users\hbk72\Desktop</directories>

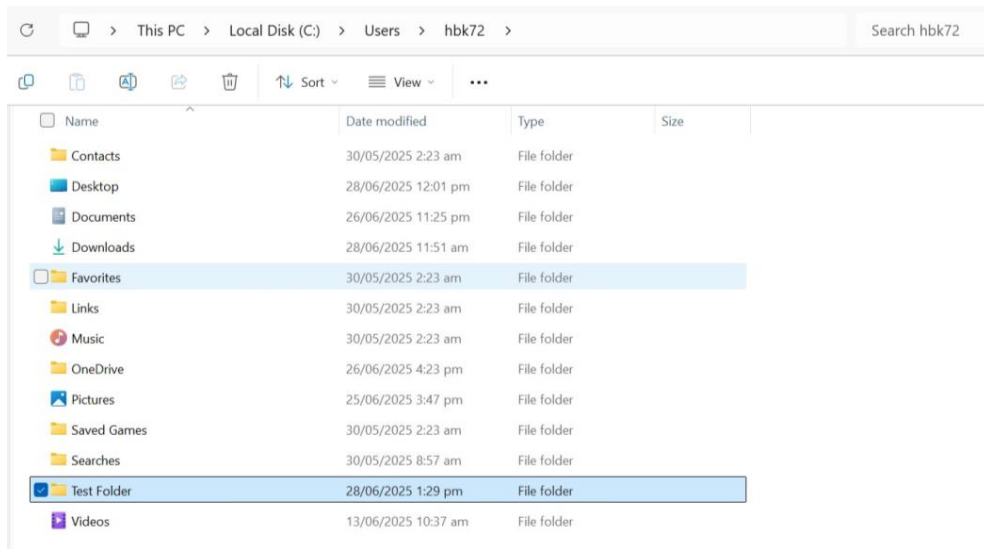
  <!-- Frequency that syscheck is executed (default every 12 hours) -->
  <frequency>43200</frequency>

  <!-- Files to be monitored -->
  <directories recursion_level="0" restrict="regedit.exe|system.ini|win.ini|cmd.exe|powershell.exe">%WINDIR%\SysNative</dir
  <directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\SysNative\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe">%WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>

  <!-- 32-bit programs -->
  <directories recursion_level="0" restrict="at.exe|ftp.exe|lsass.exe|net.exe|netsh.exe|sc.exe">%WINDIR%\System32</directo
  <directories recursion_level="0" restrict="powershell.exe">%WINDIR%\System32\WindowsPowerShell\v1.0</directories>

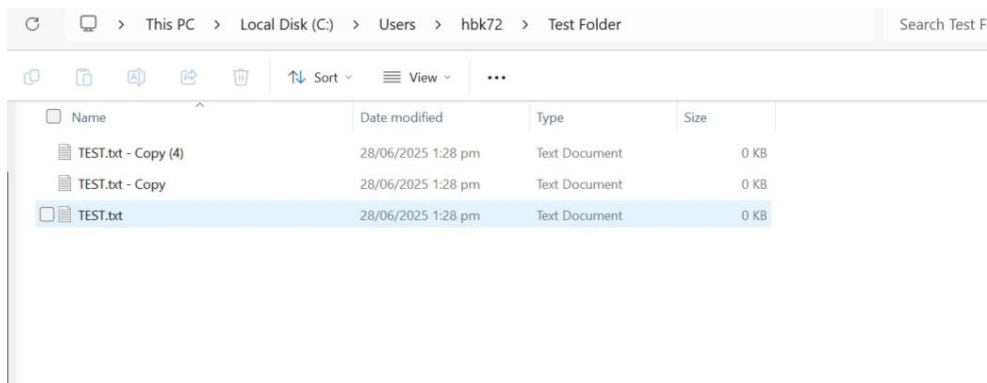
  <directories realtime="yes">%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>
  <ignore>%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup\desktop.ini</ignore>
  <ignore type="sregex">.log|.htm|.jpg|.png|.chm|.pnf|.evtx</ignore>
```

10. Make New Folder The name is Test folder

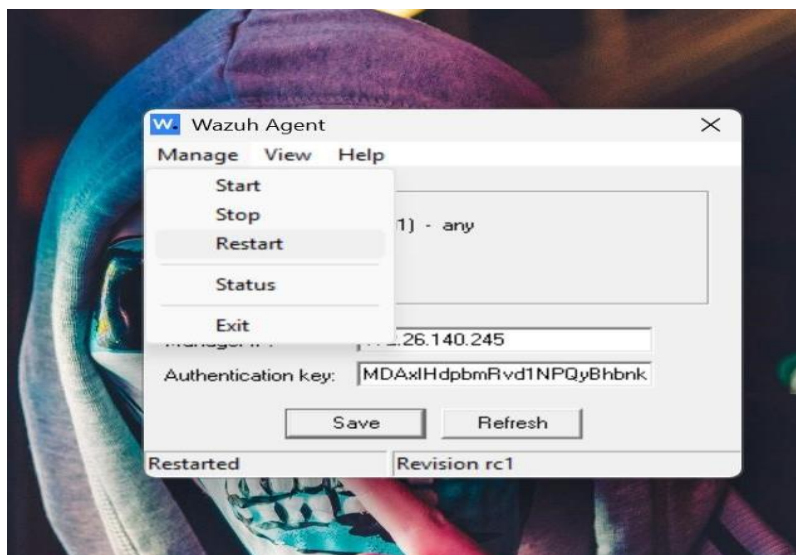


Open Test Folder

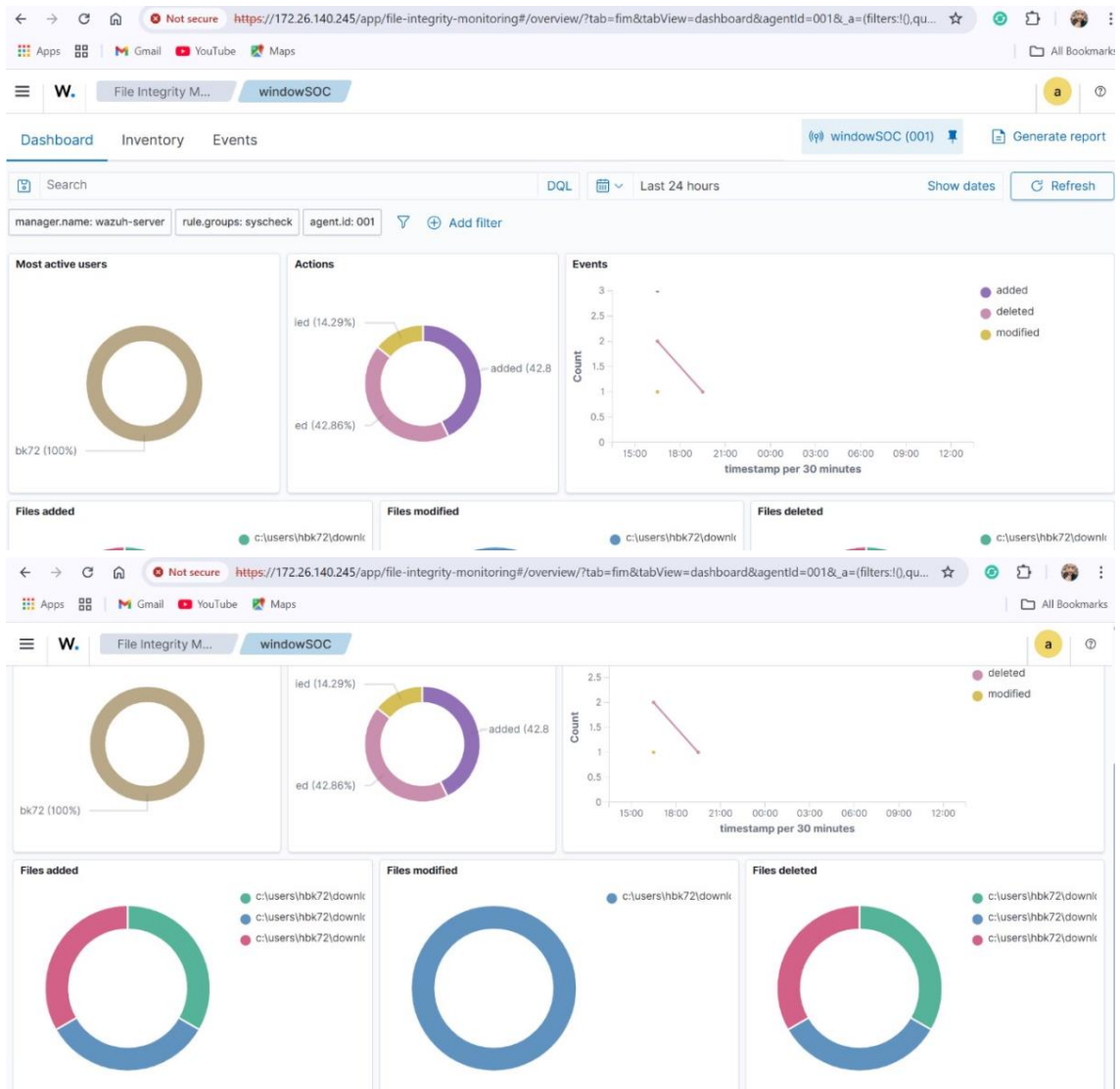
Create New some text file and deltt Some text file



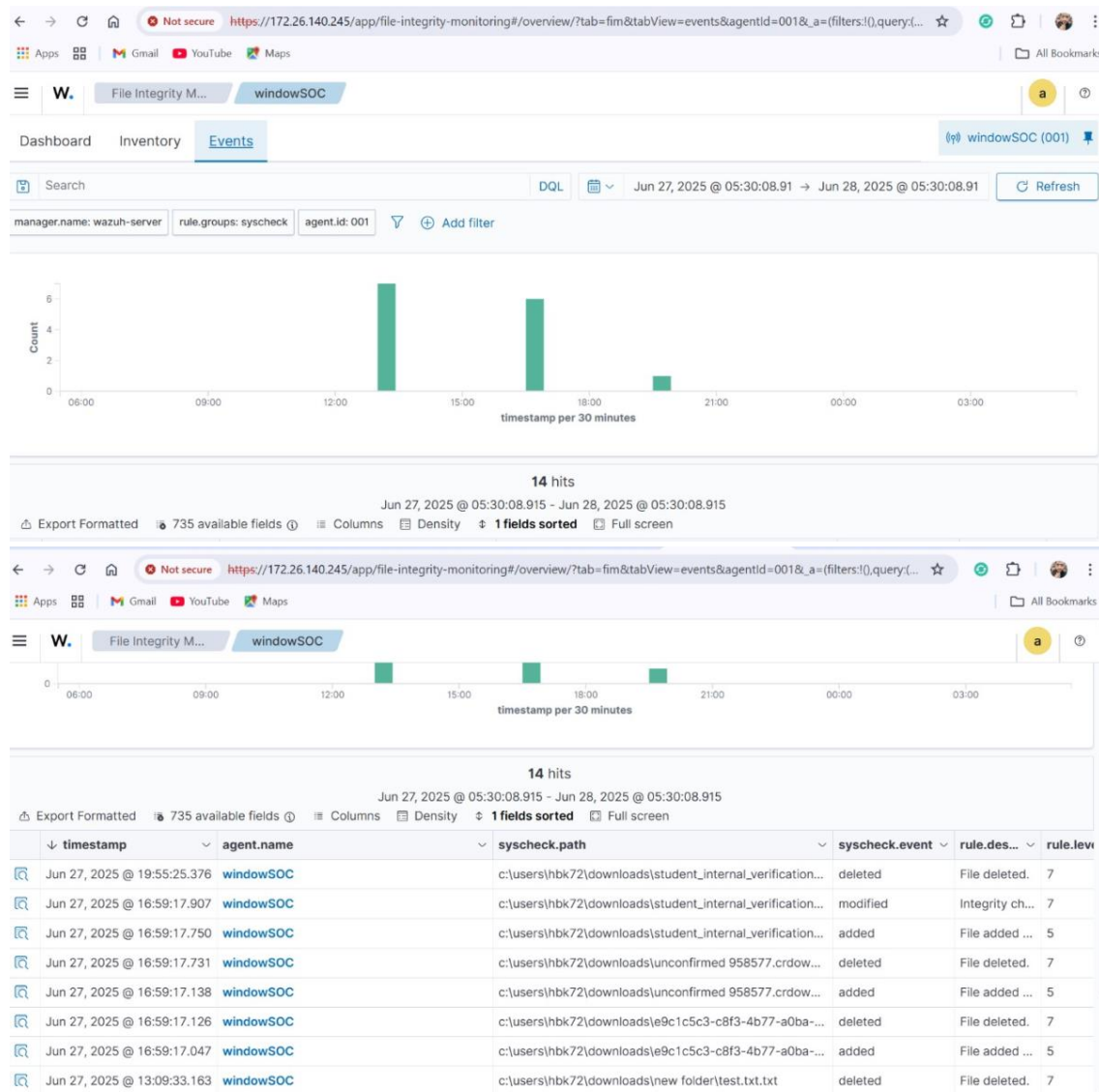
Then Restart Wazuh-agent



Open Browser and Refresh Wazuh-Dashboard



Check Alerts



11. Conclusion

File Integrity Monitoring (FIM) is a crucial layer of defense in any cybersecurity setup. By actively watching for unauthorized changes to sensitive files and directories, FIM helps detect tampering, malware activity, or insider threats in real time. With Wazuh, FIM not only logs these changes but can generate alerts, trigger automated responses, and provide compliance evidence for standards like PCI DSS, HIPAA, and ISO 27001.

In summary, FIM acts as your system's security camera for critical files — giving you visibility, accountability, and early warnings of compromise. It's essential for proactive threat detection and forensic investigations.

